Continuous coupling trajectories — run report

RUN

213414_cross-sweep_fpga-timestamp-validation
Status: completed | Started: 2026-09-29T20:34:14.390501+00:00

DIAGNOSTIC SUMMARY

Target actuator(s): phi1, phi2; sine: 0.1164 Hz; measured PAX rate: 6.9 Hz; saved RP reference rate: 381 Hz; estimated PAX samples/cycle: 59.26.
Recorded cycles: 2; bias values: phi1 target 1, phi2 target 1. Waveform tracking: correct within descriptive threshold. Rising/falling branches (>5° descriptive threshold): strong separation present. Cycle repeatability (5° descriptive threshold): nonrepeatable above threshold. Bias-dependent plane changes (>5° descriptive threshold): unresolved. Low-DoP intervals: True. These thresholds are flags, not mechanism assignments.

PURPOSE

Continuously sine-drive phi1, phi2, or both while stepping and settling the orthogonal bias; record PAX plus measured OUT1/OUT2 references.

REQUIRED OPTICAL SETUP

PAX at final output; both paths open unless prompted. Wire OUT1 to IN1 and OUT2 to IN2 as voltage references. 'both' runs the complete diagnostic for both actuators automatically.

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port
Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

RUN COMMENT

Measurement-tool calibration only: validate FPGA trigger/current timestamp anchoring on repaired dual references; two cycles per axis, raw FPGA and host fallback timing retained. Not a system characterization run.

OPERATOR SETUP NOTES

V_pi expectation only; verify channel and measured driver gain before calibration.

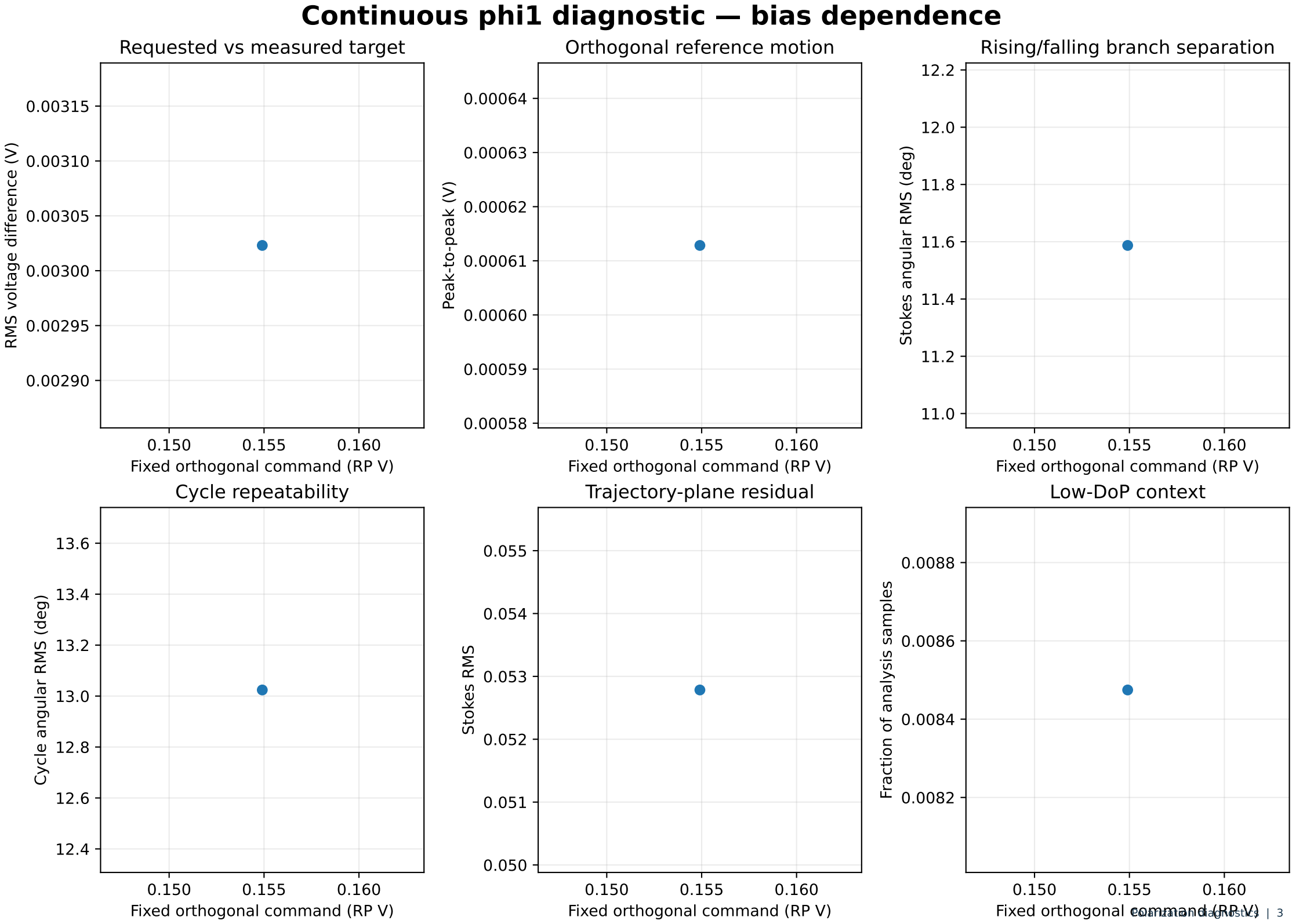
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DATA

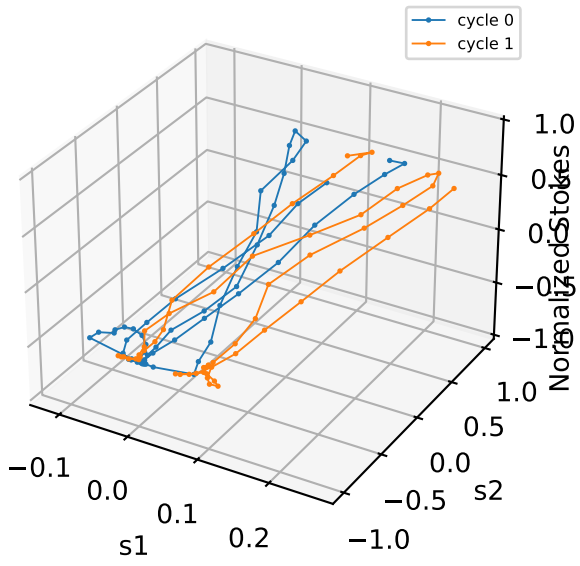
243 recorded samples; acquisition: 243
Source: data.csv; full acquisition settings: recipe.json; status: run.json

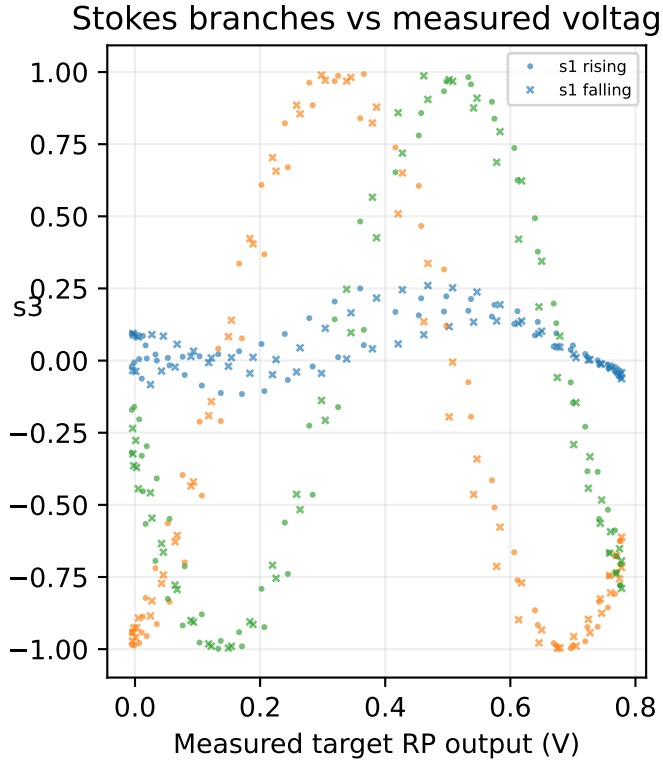
RAW DOP

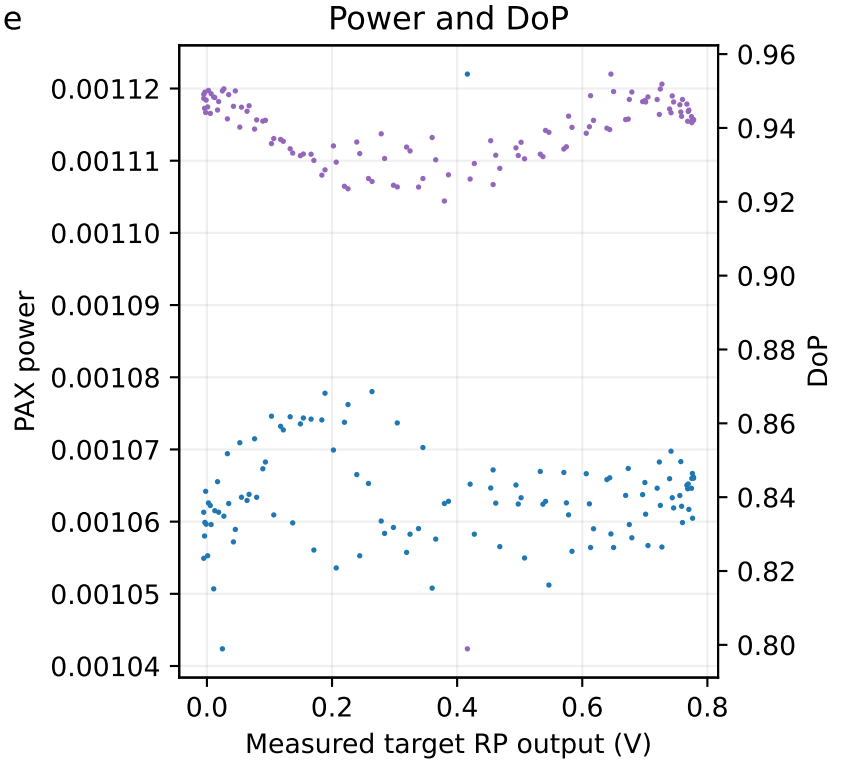
Mean 0.9231; range 0.7422 to 1.0700. 1 samples outside 0–1. Telemetry is not a new calibration.

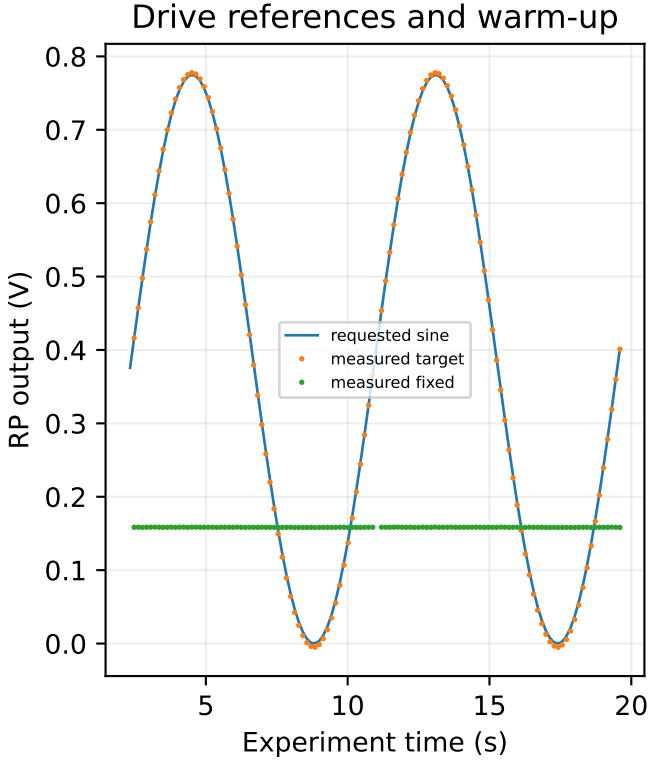

Continuous phi1 diagnostic — bias dependence
Requested vs measured target
RMS voltage difference (V)
0.00315
0.00310
0.00305
0.00300
0.00295
0.00290
0.150
0.155
0.160
Fixed orthogonal command (RP V)
Orthogonal reference motion
Peak-to-peak (V)
0.00064
0.00063
0.00062
0.00061
0.00060
0.00059
0.00058
0.150
0.155
0.160
Fixed orthogonal command (RP V)
Rising/falling branch separation
Stokes angular RMS (deg)
12.2
12.0
11.8
11.6
11.4
11.2
11.0
0.150
0.155
0.160
Fixed orthogonal command (RP V)
Cycle repeatability
Cycle angular RMS (deg)
13.6
13.4
13.2
13.0
12.8
12.6
12.4
0.150
0.155
0.160
Fixed orthogonal command (RP V)
Trajectory-plane residual
Stokes RMS
0.055
0.054
0.053
0.052
0.051
0.050
0.150
0.155
0.160
Fixed orthogonal command (RP V)
Low-DoP context
Fraction of analysis samples
0.0088
0.0086
0.0084
0.0082
0.150
0.155
0.160
Fixed orthogonal command (RP V)

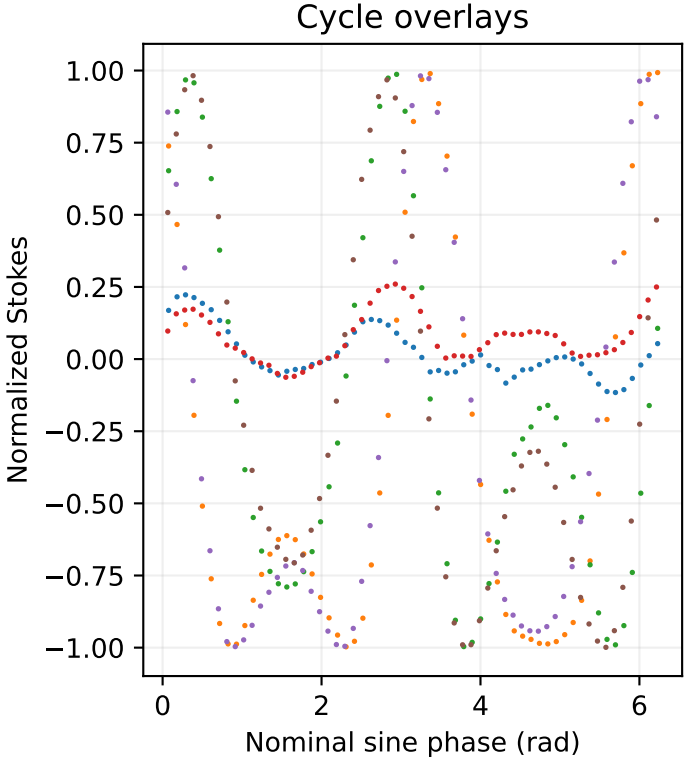
phi1 sine; fixed phi2 = 0.1549 RP V (segment 0)





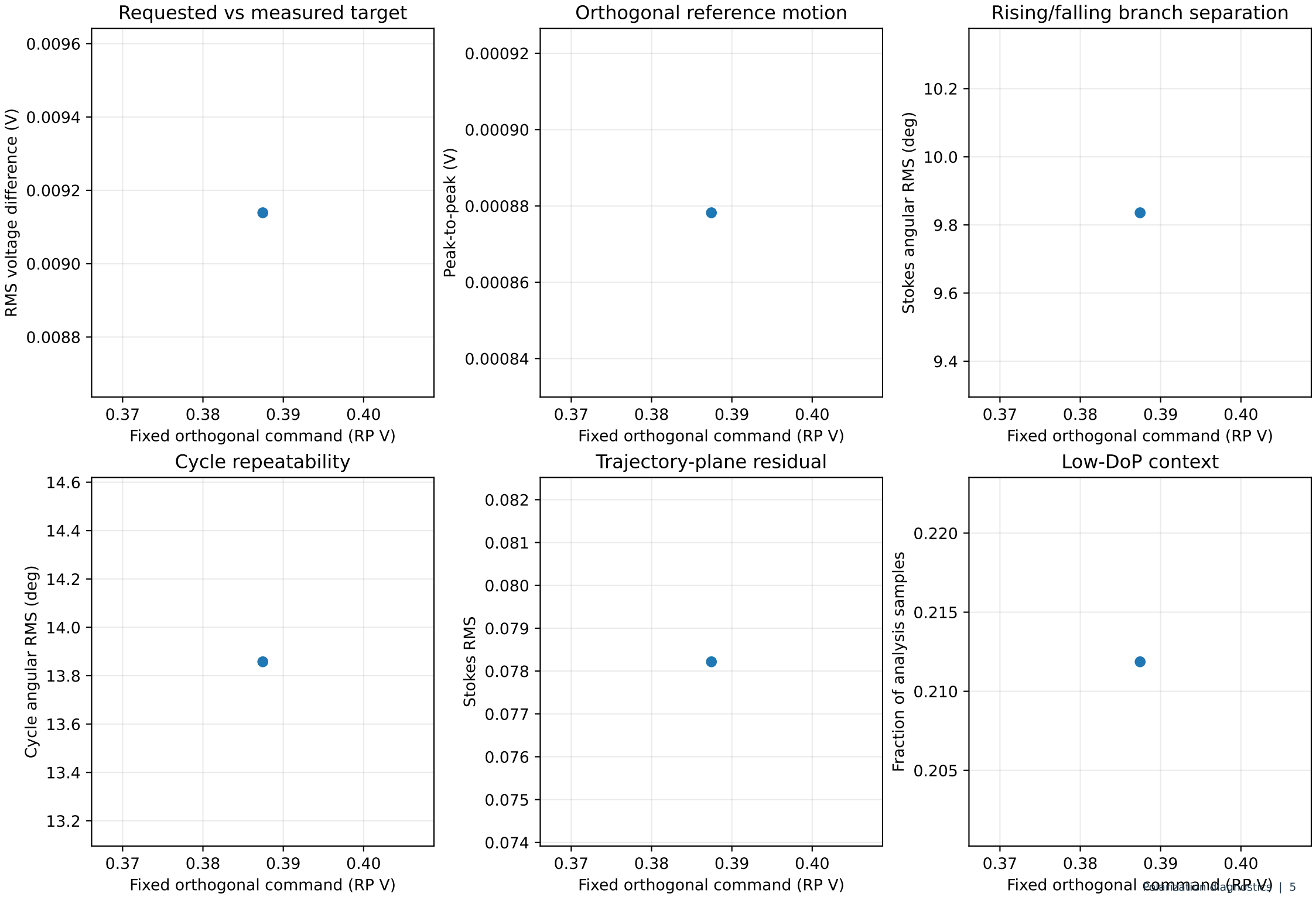




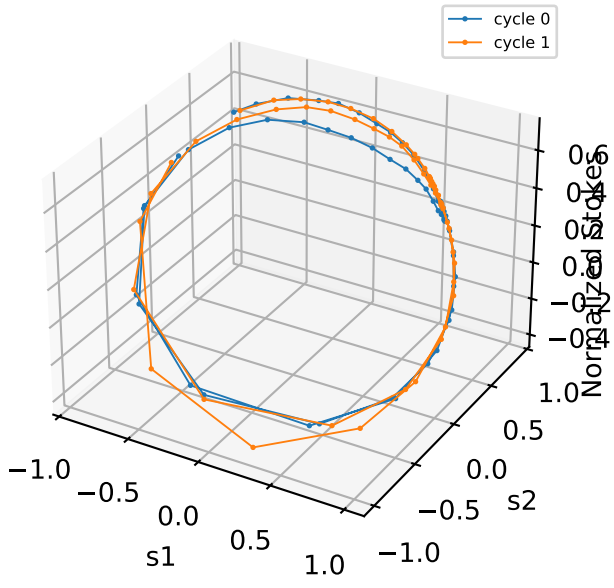


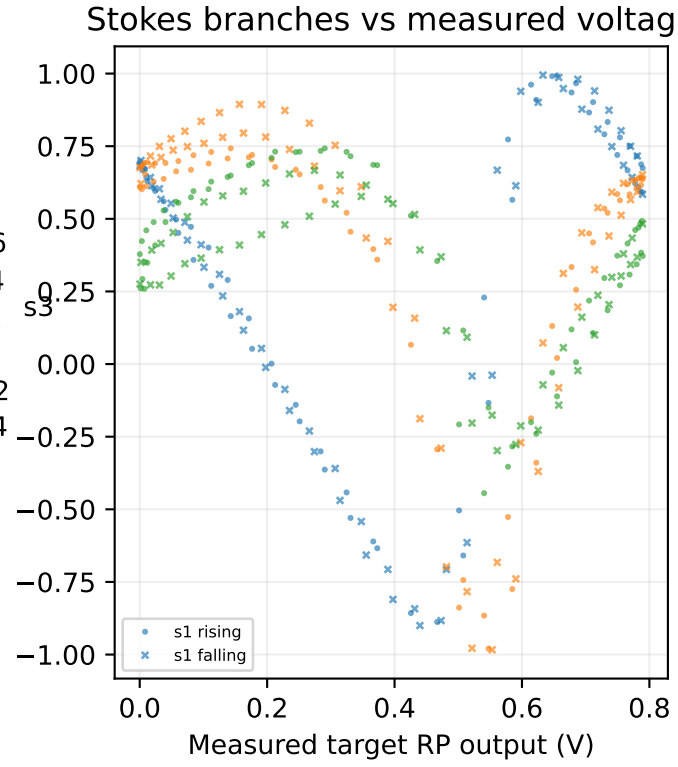
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.994, -0.016, -0.112]
Plane residual RMS: 0.05278
Rising/falling angular RMS: 11.6°
Rising/falling power RMS: 1.21e-05 PAX units
Cycle angular RMS: 13°
Cycle power RMS: 1.08e-05 PAX units
Cycle DoP RMS: 0.0196
Tracking RMS: 0.003023 V
Reference coverage: 99.2%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.99, 0.01, -0.12], residual 0.0391
Cycle 1 plane: [0.99, -0.04, -0.10], residual 0.042

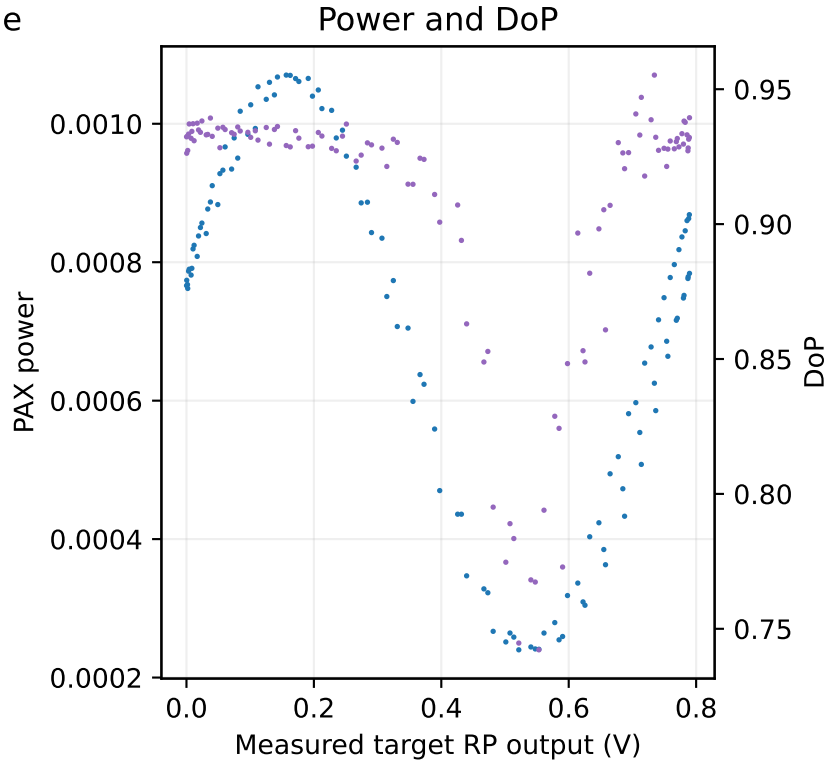
Continuous phi2 diagnostic — bias dependence

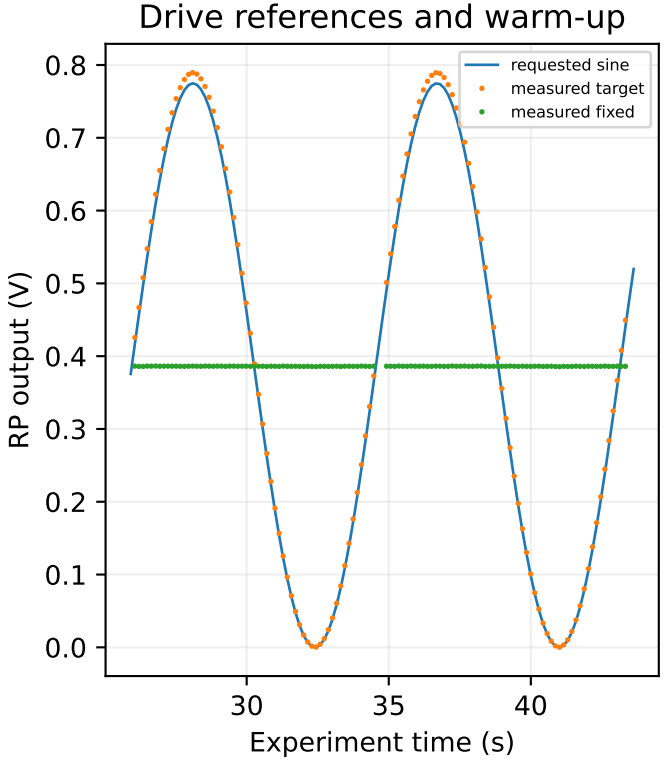


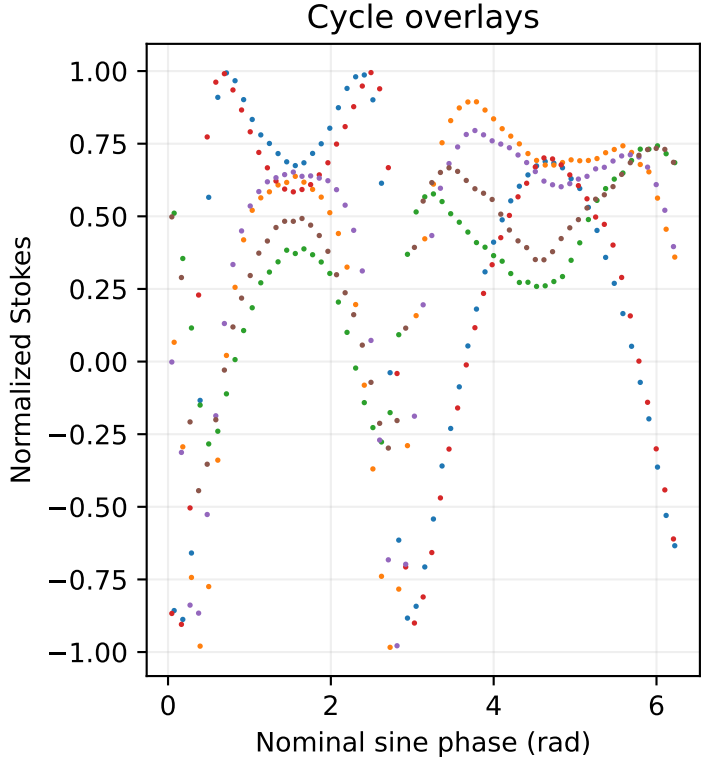
phi2 sine; fixed phi1 = 0.3875 RP V (segment 1)









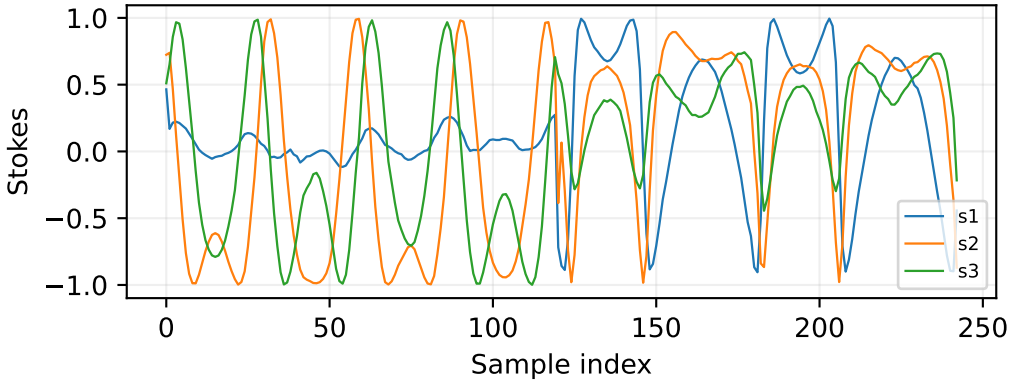


Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.266, -0.447, 0.854]
Plane residual RMS: 0.07822
Rising/falling angular RMS: 9.84°
Rising/falling power RMS: 3.36e-05 PAX units
Cycle angular RMS: 13.9°
Cycle power RMS: 5.78e-05 PAX units
Cycle DoP RMS: 0.0155
Tracking RMS: 0.009139 V
Reference coverage: 98.3%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.27, -0.39, 0.88], residual 0.082
Cycle 1 plane: [0.26, -0.50, 0.83], residual 0.0491

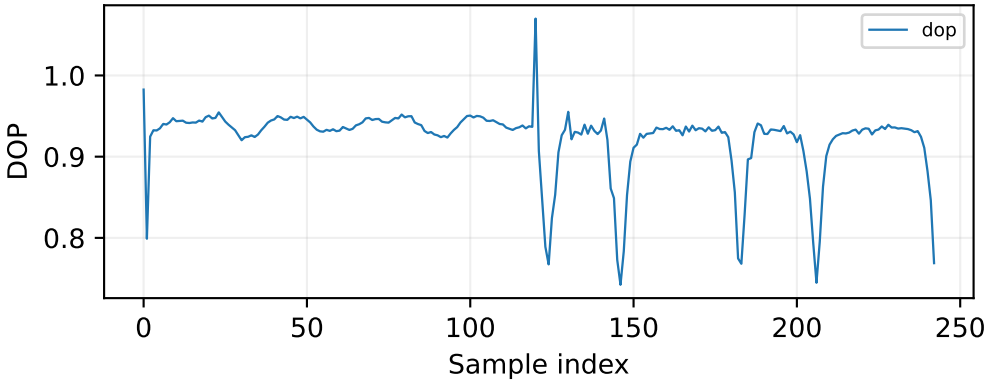
Recorded telemetry

Acquisition order preserves stage changes; no timing alignment or transfer gain is inferred.

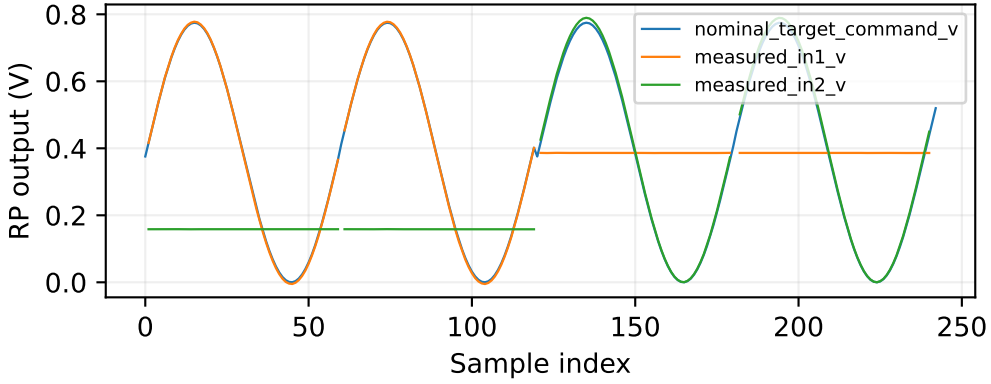
Normalized Stokes



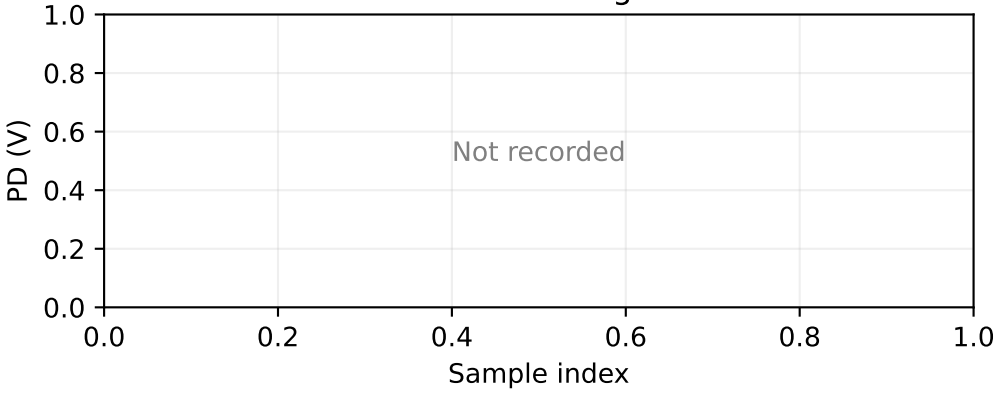
Raw degree of polarization



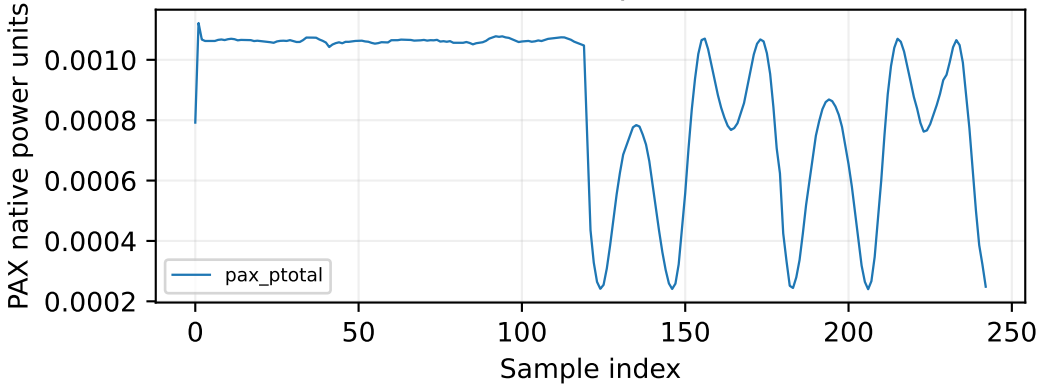
Actuator command / reference history



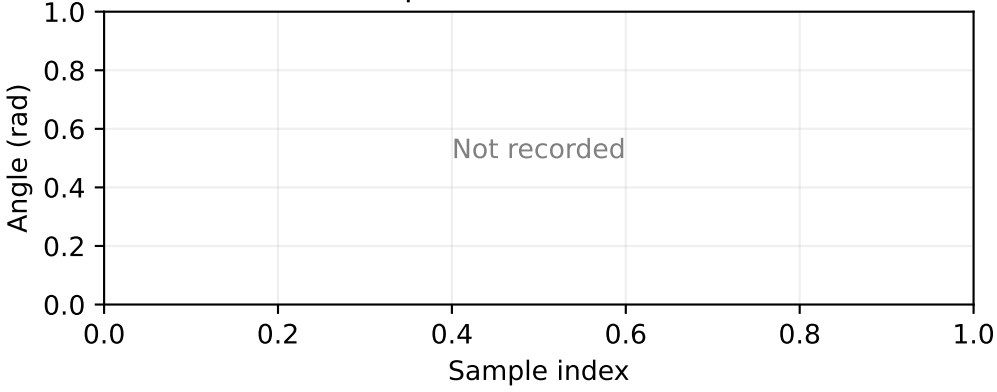
Photodiode signal



PAX total power



Sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
label	"fpga-timestamp-validation"
comment	"Measurement-tool calibration only: validate FPGA trigger/currer
actuator	"both"
report	"pdf"
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
phi1_v_lambda	13.078125
phi2_v_lambda	13.078125
phi1_actuator_volts_per_rp_volt	16.875
phi2_actuator_volts_per_rp_volt	16.875
rp_output_min_voltage	0.0
rp_output_max_voltage	1.0
cross_sweep_sine_frequency_hz	0.1
cross_sweep_sine_center_voltage	0.3875
cross_sweep_sine_amplitude_voltage	0.3875
cross_sweep_warmup_cycles	0
cross_sweep_recorded_cycles	2
cross_sweep_phi1_bias_voltages	[0.3875]
cross_sweep_phi2_bias_voltages	[0.155]
cross_sweep_settle_s	1.0
cross_sweep_bias_ramp_s	1.0
cross_sweep_bias_ramp_updates_per_s	50.0
cross_sweep_reference_sample_rate_hz	400.0

Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
cross_sweep_scope_block_s	8.0
cross_sweep_min_pax_samples_per_cycle	20.0
cross_sweep_pax_latency_s	0.12